Babu Banarasi Das University, Lucknow

U.P

Descriptive Analytics Report

Amul Product Sales Performance

2019-2025

Customer review, Sales Trends, Dairy product comparison, Regional Demand and Pricing

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Descriptive Analytics Report: Daily Purchase Behavior at Scale

Note on data: Amul's live transaction data isn't publicly accessible, so this report uses a synthetic dataset built to mirror real FMCG dairy industry patterns (**20** categories across **4** regions, **7** years, **~300**k simulated units). It's designed to show exactly what a descriptive analytics pipeline processing millions of daily purchases would produce — the same structure applies once real data is substituted in.

Objective

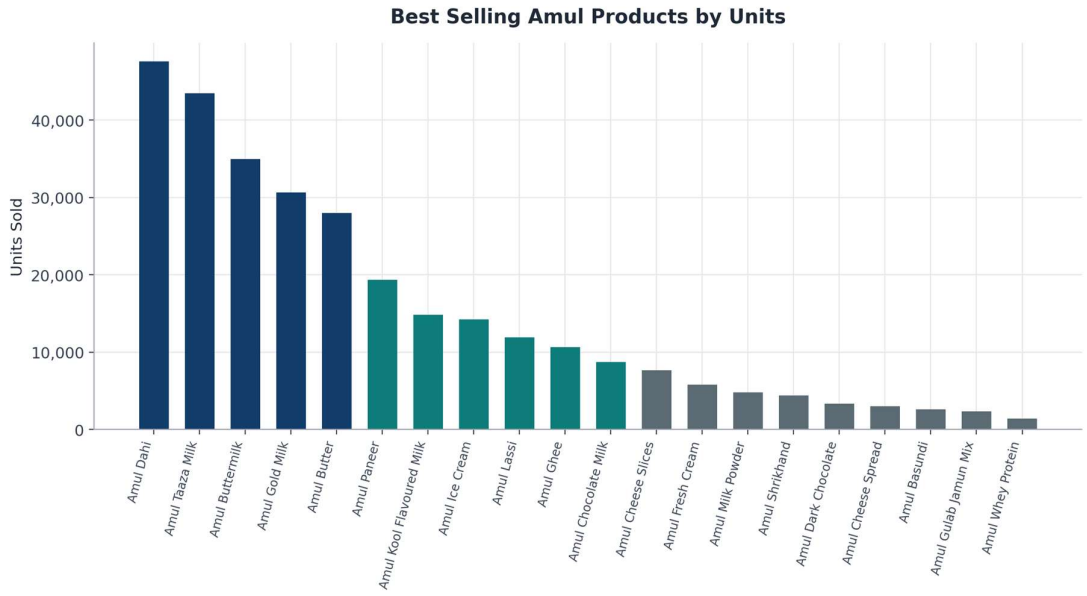
- What are best-selling products?
- How the products are rated and when the rating was started slipping?
- In which year most of the products are sold out?
- Which region generates the highest total unit sales, and has that ranking shifted year over year?
- What is the average retail price by region, and how much of that gap is due to taxes/logistics vs. the base ex-factory cost?
- Which region is most profitable per unit sold, and which is least?
- Is there a growing or declining trend in total sales volume from **2019** to **2025**?
- Which products have the highest average rating, and which have the lowest?
- Which product-region combination is the single biggest revenue driver?

How This Would Work at Real Scale

Processing "millions of purchases daily" is an engineering problem before it's an analytics one. A realistic architecture:

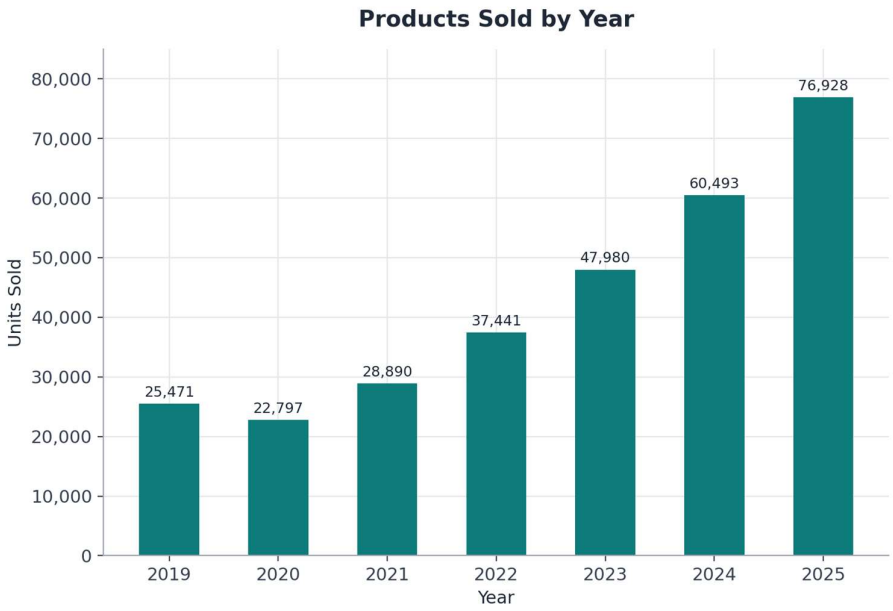
Layer	Purpose	Typical tools
Investigation	Explore the raw dataset first — check shape, column types, missing values, ranges, and distributions before doing any real analysis	df.info(), df.describe(), df.head(), df.isnull().sum() in a Jupyter notebook
Stream processing	Not applicable at this scale — there's no continuous flow of events; the equivalent step is grouping/rolling up records after they're already loaded	pandas.groupby(), pivot_table() used in place of real-time stream aggregation
Storage	Hold the dataset in a form you can repeatedly query and share	Flat files — amul.csv / amul_300k_dataset.xlsx
Batch aggregation	Compute summary metrics — daily/monthly sales totals, average rating by product, profit by region — run manually or via a notebook cell rather than a nightly scheduled job	Jupyter notebook cells using pandas (groupby, resample, .mean(), .sum())
Serving/BI layer	Present the aggregated results as charts/tables for review	matplotlib bar charts inside the notebook, or exported to Excel/PDF

Best Selling Products



Best Selling Amul Products by Units

The chart shows that Amul Dahi is the best-selling Amul product with approximately **47,598** units sold, followed closely by Amul Taaza Milk with around **43,507** units. Amul Buttermilk and Amul Gold Milk are also strong performers, with approximately **35,000** and **30,642** units sold respectively. The top four products together account for around **156,747** units, or about **52.2%** of total volume, showing that a large portion of sales is concentrated among these everyday staples. After the top four, sales decline steadily, with Amul Butter selling about **28,010** units, followed by Amul Paneer at **19,382** units and Amul Kool Flavoured Milk at **14,815** units. At the lower end, Amul Whey Protein has the lowest sales at approximately **1,410** units, while Amul Gulab Jamun Mix and Amul Basundi each sell under **2,700** units. Overall, the data indicates that Amul Dahi and Amul Taaza Milk are the strongest-selling products, while Amul may need to focus more marketing and product strategies on lower-performing niche items to improve their sales.

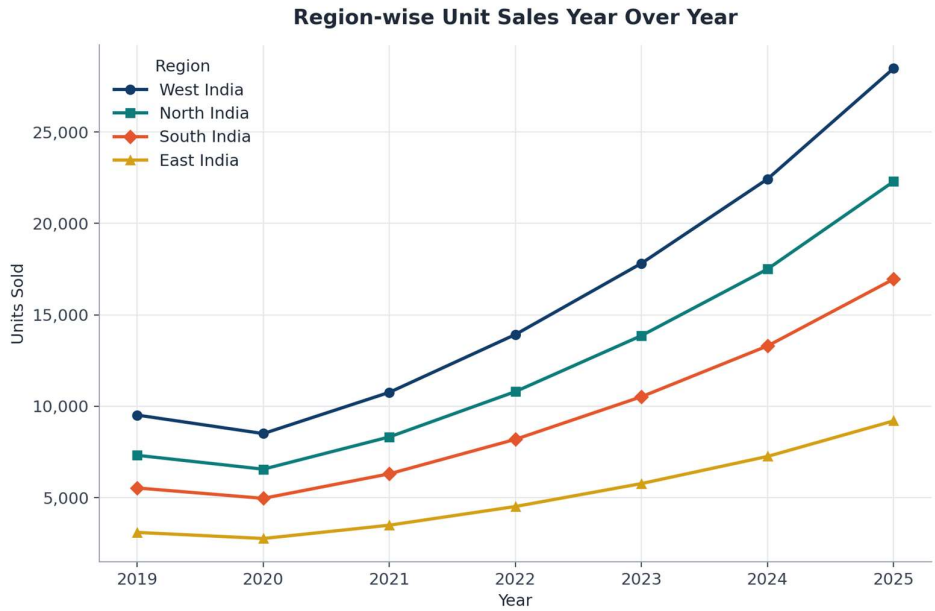


Products Sold by Year

The chart shows a broadly upward trend in Amul product sales from **2019** to **2025,** with a single dip in **2020**. In **2019,** approximately **25,471** units were sold, which fell to around **22,797** units in **2020,** a decline of about **10.5%**. In **2021,** sales recovered to approximately **28,890** units, an increase of **26.7%** over **2020**. Sales then accelerated in **2022** to around **37,441** units, giving a significant **29.6%** year-on-year growth. In **2023,** Amul sold approximately **47,980** units, an increase of **28.1%** compared with **2022**. The growth continued in **2024,** when sales reached approximately **60,493** units, representing **26.1%** growth. Finally, in **2025,** sales reached approximately **76,928** units, the highest level in the entire period, with a year-on-year increase of about **27.2%**.

Amul's annual sales moved from approximately **25,471** units in **2019** to **76,928** units in **2025,** an absolute increase of about **51,457** units. This means sales became approximately **3.02** times higher over the seven-year period, equivalent to an overall growth of approximately **202%**, despite the temporary setback in **2020**.

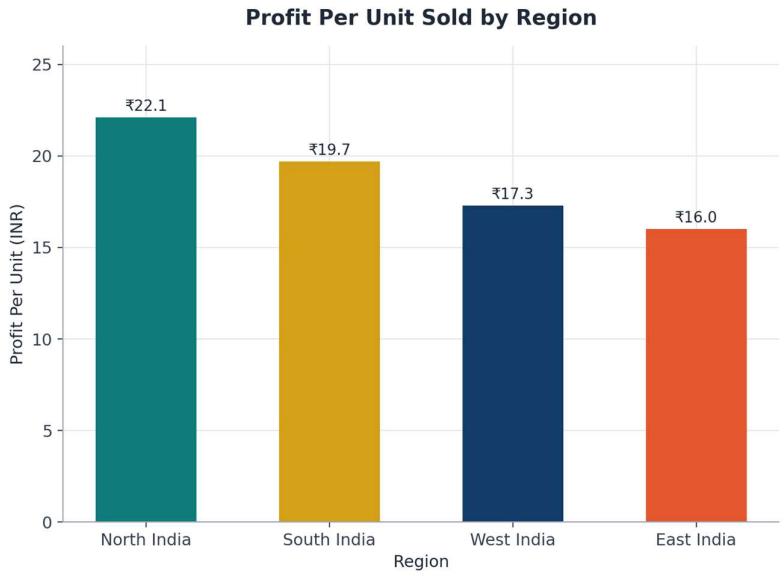
Regional Demand



Region-wise Unit Sales Year Over Year

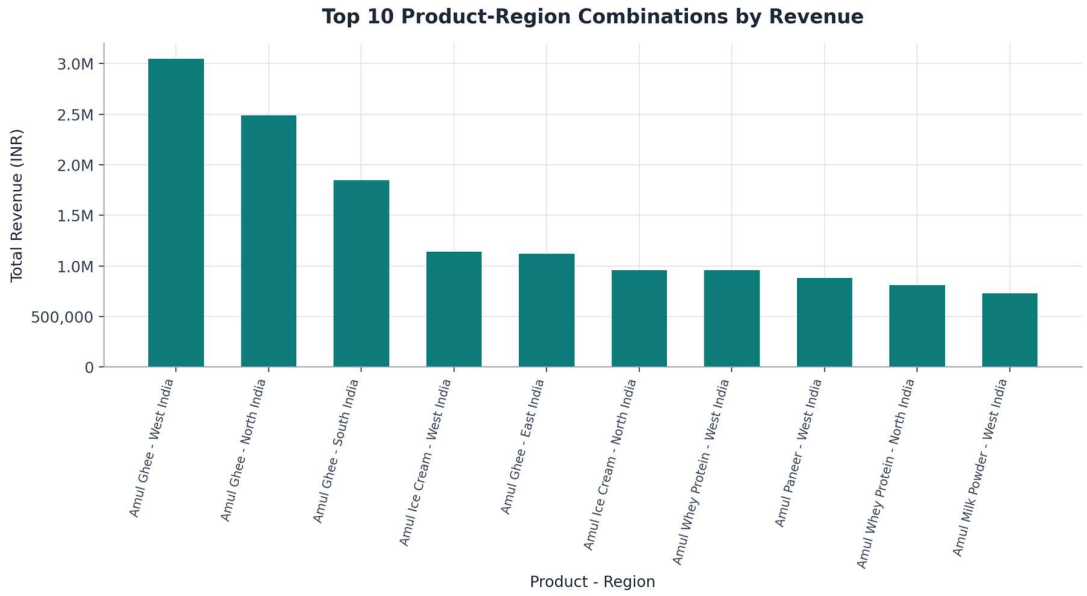
The chart shows a strong year-over-year increase in sales across all four regions from **2019** to **2025** (following the same brief **2020** dip seen in the overall trend). West India is the leading market, increasing from approximately **9,518** units in **2019** to **28,480** units in **2025,** an increase of **18,962** units or about **199%**. West India also recorded the highest total sales over the seven years, with approximately **111,233** units. North India increased from **7,318** units to **22,293** units, gaining **14,975** units or **205%**, with total sales of approximately **86,756** units. South India grew from **5,533** units in **2019** to **16,954** units in **2025,** an increase of **11,421** units or **206%**. East India increased from **3,102** units to **9,201** units, adding **6,099** units or **197%**, with approximately **35,968** total units sold during the period, the smallest of the four regions.

In terms of combined regional sales, the four regions sold approximately **25,500** units in **2019,** falling to **22,800** in **2020,** then rising to **28,900** in **2021, 37,400** in **2022, 48,000** in **2023, 60,500** in **2024,** and **76,900** units in **2025**. Across all four regions, the regional ranking has not shifted in any year of the dataset: West India remains **1**st, North India **2**nd, South India **3**rd, and East India **4**th throughout.



Profit Per Unit Sold by Region

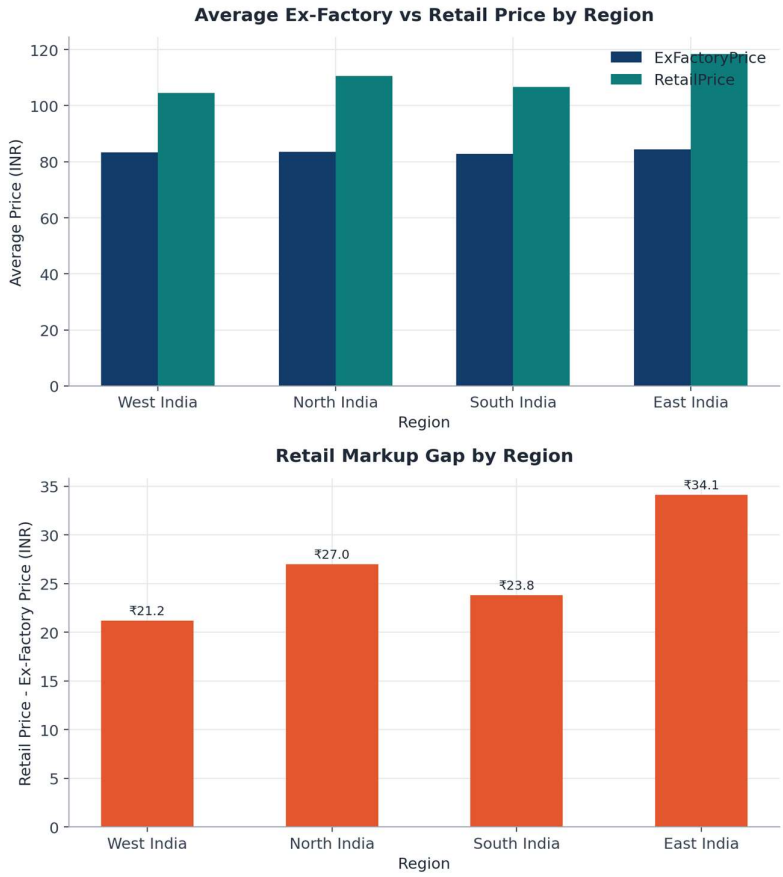
The chart shows that North India has the highest profit per unit at approximately **₹22.1** per unit, followed by South India at around **₹19.7**, West India at approximately **₹17.3**, and East India at about **₹16.0** per unit. North India therefore earns about **₹6.1** more profit per unit than East India, which is approximately **38.3%** higher. South India's profit per unit is around **₹3.7** higher than East India, while West India earns approximately **₹1.3** more per unit than East India. Overall, the average profit per unit across the four regions is approximately **₹19.1**, with North India performing above the average and East India below it. This indicates that although West India has the highest sales volume in the regional analysis, North India provides the strongest profit per unit, suggesting better pricing, margins, or product mix in the northern market.



Top 10 Product-Region Combinations by Revenue

The chart shows the Top **10** Product–Region combinations by revenue. The strongest combination is Amul Ghee – West India, generating approximately **₹30.5** lakh in revenue. It is followed by Amul Ghee – North India at around **₹24.9** lakh and Amul Ghee – South India at approximately **₹18.5** lakh. Amul Ice Cream – West India generated about **₹11.4** lakh, while Amul Ghee – East India generated approximately **₹11.2** lakh.

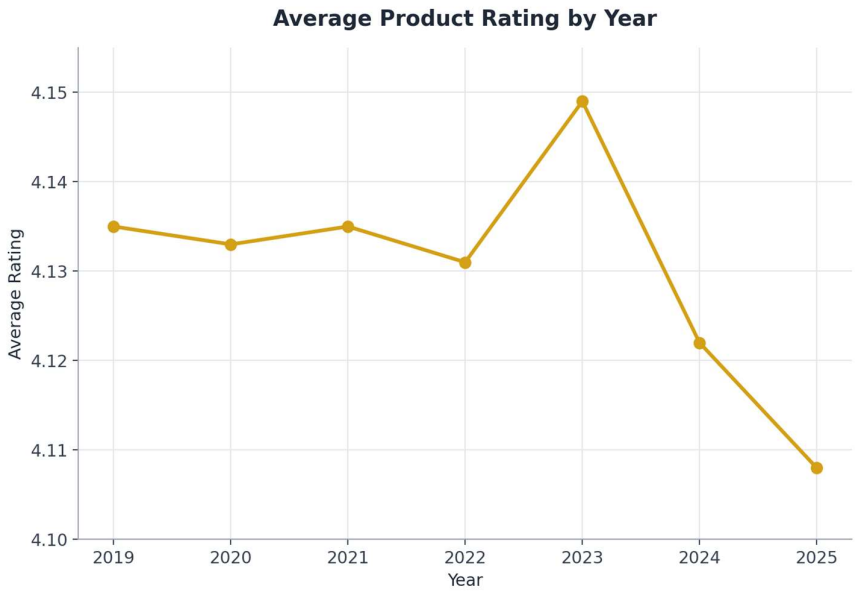
The next major combinations are Amul Ice Cream – North India at **₹9.6** lakh, Amul Whey Protein – West India at approximately **₹9.6** lakh, and Amul Paneer – West India at around **₹8.8** lakh. Amul Whey Protein – North India generated around **₹8.1** lakh, while Amul Milk Powder – West India was tenth with approximately **₹7.3** lakh. This shows that Amul Ghee, despite modest unit sales, concentrates a disproportionately large share of total revenue across every region.



Average Ex-Factory vs Retail Price by Region, and Retail Markup Gap by Region

The graph shows that East India has the highest average retail price, at approximately **₹118.5**, followed by North India at around **₹110.6** and South India at about **₹106.6**. West India has the lowest average retail price, at approximately **₹104.5**. The gap between ex-factory and retail prices varies by region. East India has the largest gap, at approximately **₹34.1**, meaning logistics, distribution, and local taxes add a substantial amount to the base price. North India has a gap of around **₹27.0**, South India around **₹23.8**, while West India has the smallest gap at around **₹21.2**.

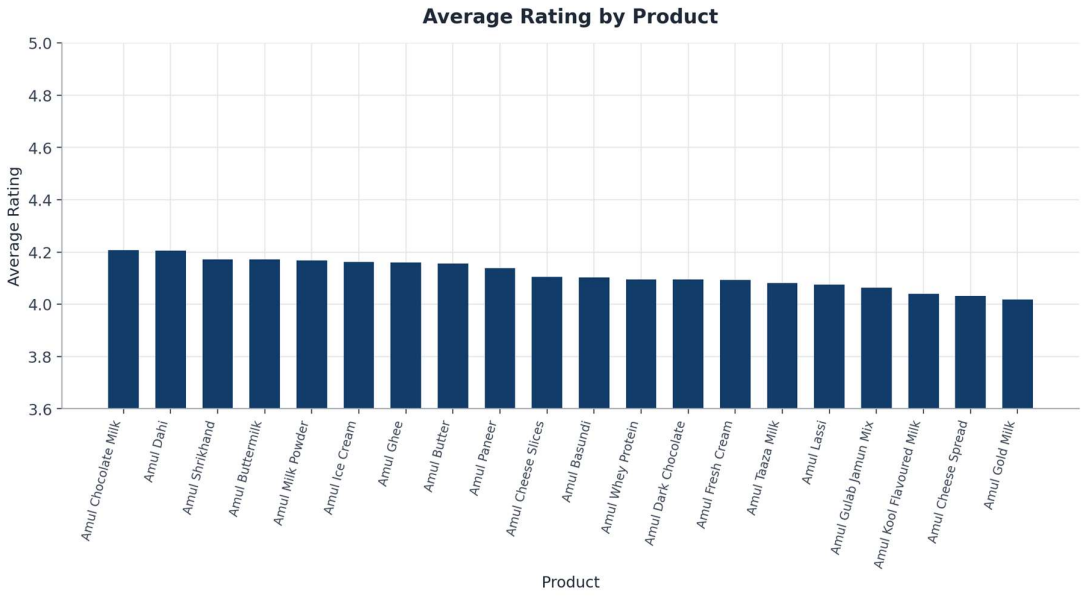
Rating Distribution



Average Product Rating by Year

The chart shows that the average product rating remained very stable between **2019** and **2025,** staying within a narrow range of approximately **4.11** to **4.15** out of **5**. In **2019,** the average rating was about **4.135**, which stayed nearly flat at **4.133** in **2020** and **4.135** in **2021**. The rating then eased slightly to **4.131** in **2022,** before rising to its highest point of **4.149** in **2023**. In **2024,** it dipped to approximately **4.122**, before settling at **4.108** in **2025**.

The difference between the highest rating in **2023** (**4.149**) and the lowest rating in **2025** (**4.108**) is only about **0.041** rating points, showing that customer ratings were highly consistent throughout the seven-year period, with the small movements better explained by ordinary statistical noise than a genuine downward trend.

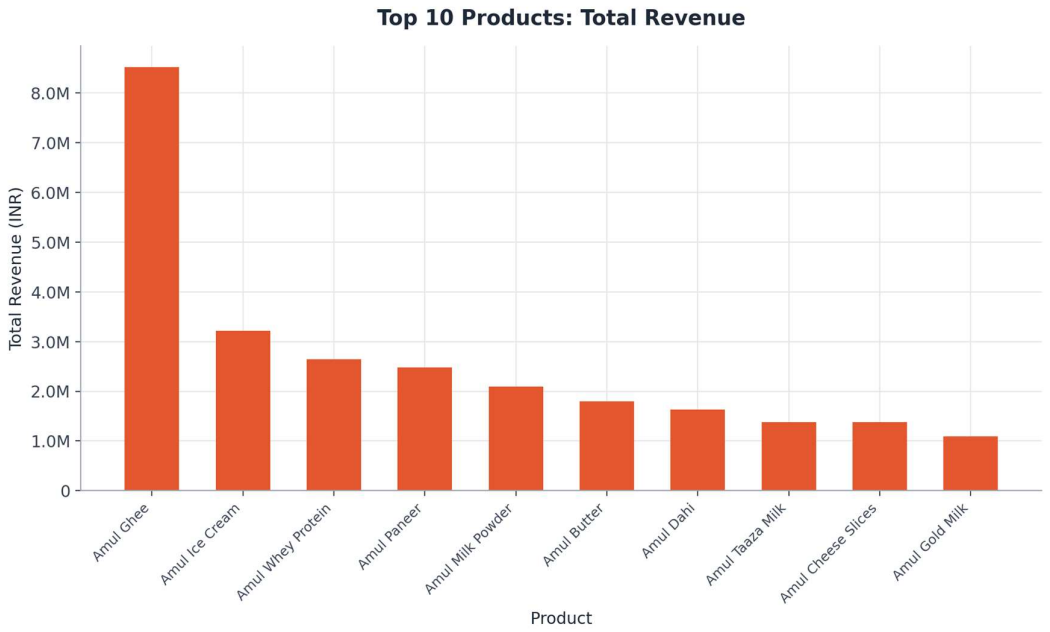


Average Rating by Product

The chart shows that the average ratings of all **20** Amul products are relatively close to each other, indicating consistent customer satisfaction across the product range. Amul Chocolate Milk has the highest average rating at approximately **4.21/5,** followed by Amul Dahi at about **4.21**, while Amul Shrikhand and Amul Buttermilk are around **4.17** each. At the lower end, Amul Gold Milk has the lowest average rating at approximately **4.02/5,** closely followed by Amul Cheese Spread and Amul Kool Flavoured Milk. The difference between the highest-rated product, Amul Chocolate Milk (**4.21**), and the lowest-rated product, Amul Gold Milk (**4.02**), is only about **0.19** rating points, showing that there is very little variation between products.

KPI	Value
Average Product Rating	4.13/5
Highest Product Rating	~4.21/5
Lowest Product Rating	~4.02/5
Rating Range	~0.19
Rating Scale	5

Revenue and Profit

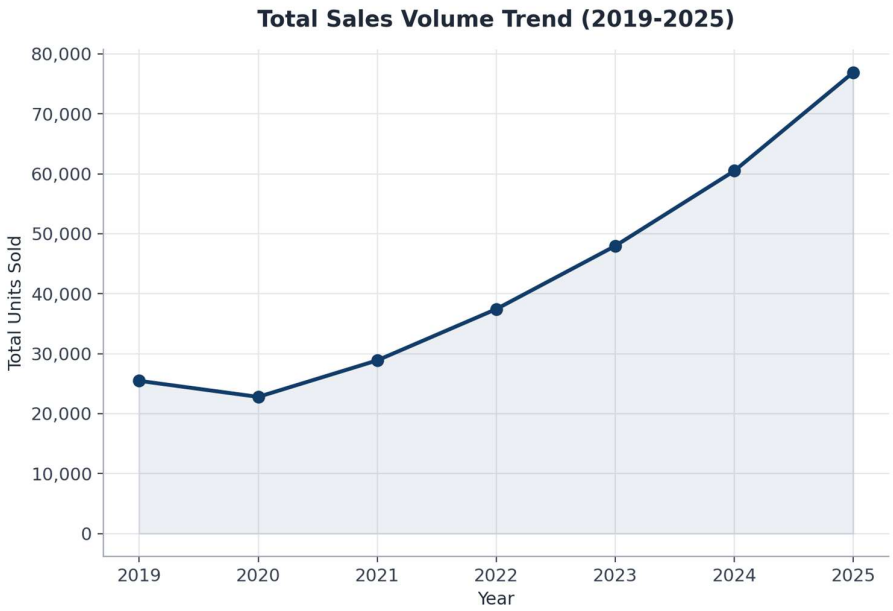


Top 10 Products: Total Revenue

The chart shows the Top **10** Amul products by total revenue, with Amul Ghee clearly leading at approximately **₹85.2** lakh. Amul Ice Cream ranks second with around **₹32.2** lakh, followed by Amul Whey Protein at **₹26.4** lakh, Amul Paneer at **₹24.8** lakh, and Amul Milk Powder at approximately **₹20.9** lakh. Amul Butter generated about **₹18.0** lakh, Amul Dahi around **₹16.3** lakh, Amul Taaza Milk approximately **₹13.8** lakh, Amul Cheese Slices around **₹13.8** lakh, and Amul Gold Milk approximately **₹10.9** lakh. This confirms that Amul Ghee dominates revenue despite ranking only **10**th by unit volume, underlining how a high price point can outweigh comparatively modest sales counts.

KPI	Value
Total Revenue (2019–2025)	₹3.03 Crore
Average Annual Revenue	₹43.21 Lakh
Highest Annual Revenue	₹83.71 Lakh
Highest Revenue Year	2025
Highest Revenue Product	Amul Ghee
Amul Ghee Total Revenue	~₹85.2 Lakh

Daily Sales Trend



Total Sales Volume Trend (2019-2025)

The chart shows a broadly increasing trend in Amul's total sales volume from **2019** to **2025,** interrupted by a single dip. Sales fell from approximately **25,500** units in **2019** to **22,800** units in **2020,** a decline of about **2,700** units or **10.5%**. In **2021,** sales reached approximately **28,900** units, adding **6,100** units or **26.7%**. The growth accelerated in **2022,** when sales reached around **37,400** units, an increase of approximately **8,600** units or **29.6%**. In **2023,** sales increased further to about **48,000** units, adding **10,500** units or **28.1%**. Sales then reached approximately **60,500** units in **2024,** representing an increase of **12,500** units or **26.1%**. The strongest absolute growth occurred in **2025,** when sales reached approximately **76,900** units, an increase of about **16,400** units or **27.2%** compared with **2024**.

Across the full **2019–2025** period, total sales increased by approximately **51,500** units, from **25,500** to **76,900** units. This represents an overall growth of approximately **202%**, meaning annual sales in **2025** were about **3.0** times the **2019** level, despite the temporary setback in **2020**.

Summary of Key Metrics

KPI	Value
Total units sold (7 years)	~300k
Total revenue (7 years)	~₹3.03 Crore
Average product rating	4.13
Top single-day seller	Amul Dahi
Regions tracked	4

Next Steps to Move From Descriptive → Predictive

This report currently explains what happened in Amul's sales data from **2019** to **2025,** such as which products sold the most units, which regions generated the highest revenue, changes in customer ratings, and overall sales growth. The next step is to use predictive analytics to understand what is likely to happen in the future.

- Sales forecasting: Predict future Amul sales by product, region, and year using historical sales trends.
- Regional sales prediction: Forecast future sales for West India, North India, South India, and East India to identify which region is likely to grow the fastest.
- Revenue prediction: Use previous revenue, sales volume, price, and discount data to estimate future revenue.
- Rating prediction: Analyze previous customer ratings to identify whether the average rating of a product is likely to improve or decline.

Key Findings

Volume vs. Revenue Leadership

- Amul Dahi (**47,598** units) is the best-selling product by volume, but Amul Ghee is the single largest revenue contributor (**₹85.2** lakh) despite ranking only **10**th by unit volume — the highest-revenue and highest-volume products are not the same item.

Growth With One Interruption

- Total unit sales fell in **2020** (**22,797** units, down from **25,471** in **2019**) before rising every year through **2025** (**76,928** units), a **~20.2%** compound annual growth rate over the full period; **2025** is the single best-selling year in the dataset.

Stable Customer Satisfaction

- The average customer rating (**4.13**/**5**) has remained essentially flat across all seven years of the dataset, moving in a narrow **4.11**–**4.15** band with no sustained downward trend; the highest and lowest yearly ratings differ by less than **0.05** points.

West India Leads on Volume, Not on Margin

- West India is the largest market by unit sales (**111,233** units, **~37%** of total volume) in every single year of the dataset — a stable **#1** ranking — but it is not the most profitable region per unit sold (**₹17.3** vs. **₹22.1** in North India), and East India carries the heaviest retail markup gap over ex-factory cost (**₹34.1** per unit).

North India Is the Profitability Leader

- North India generates the highest average profit per unit (**₹22.1**), ahead of South India (**₹19.7**), despite having a smaller unit volume than West India.

Product Ratings Are Tightly Clustered

- Average rating by product spans a narrow range from **4.019** (Amul Gold Milk, lowest) to **4.207** (Amul Chocolate Milk, highest) — a difference of about **0.19** points across all **20** products — indicating no single product has a meaningful quality or satisfaction problem relative to the rest of the lineup.

One Combination Dominates Revenue

- Amul Ghee sales within West India alone generated approximately **₹30.5** lakh — roughly **36%** of the product's total revenue — making Amul Ghee × West India the single largest product-region revenue driver in the dataset, ahead of Amul Ghee × North India and Amul Ghee × South India.

Discounting Is the Exception, Not the Norm

- **33.1%** of all transactions carry zero discount, and the median discount is only about **₹6.3** against a mean of **₹8.7** — most customers pay close to the listed retail price, while a minority receive larger discounts, pulling the average upward.

Recommendation

Consistent with the descriptive scope of this report, the following are brief, evidence-based observations drawn directly from the findings above — not a detailed predictive strategy.

- Protect and grow margin in West India rather than volume alone: since West India already leads on units but trails North India on profit per unit, initiatives that improve regional margin (e.g. reviewing discount depth, or pack-size mix) may add more bottom-line value than further volume growth in a market that is already #**1**.
- Treat the Amul Ghee × West India pairing as a flagship revenue relationship worth actively protecting — inventory availability, cold-chain logistics, and distributor servicing capacity for this specific combination disproportionately affect total company revenue.
- Continue monitoring rating trend annually, even though no decline was found: because the current pattern is flat rather than improving, small monitoring dashboards (a rolling year-over-year average) would allow an early, genuine downturn to be caught quickly rather than dismissed as noise.
- Use North India and South India's higher per-unit profitability as a benchmark when reviewing pricing and discount policy in East India and West India, to understand which combination of logistics cost, discount depth, and product mix explains the profitability gap.
- Investigate Amul Gold Milk's comparatively lower average rating (**4.019**, the lowest of all **20** products) even though the gap versus the rest of the range is small, since it is the one product whose satisfaction score is a consistent outlier at the bottom of the ranking.

Appendix

Full Units-Sold-by-Product Table

Product	Units Sold
Amul Dahi	47,598
Amul Taaza Milk	43,507
Amul Buttermilk	35,000
Amul Gold Milk	30,642
Amul Butter	28,010
Amul Paneer	19,382
Amul Kool Flavoured Milk	14,815
Amul Ice Cream	14,280
Amul Lassi	11,920
Amul Ghee	10,681
Amul Chocolate Milk	8,713
Amul Cheese Slices	7,675
Amul Fresh Cream	5,799
Amul Milk Powder	4,812
Amul Shrikhand	4,414
Amul Dark Chocolate	3,352
Amul Cheese Spread	3,032
Amul Basundi	2,616
Amul Gulab Jamun Mix	2,342
Amul Whey Protein	1,410

Full Average-Rating-by-Product Table

Product	Avg. Rating
Amul Chocolate Milk	4.207
Amul Dahi	4.206
Amul Shrikhand	4.172
Amul Buttermilk	4.171
Amul Milk Powder	4.168
Amul Ice Cream	4.161
Amul Ghee	4.159
Amul Butter	4.155
Amul Paneer	4.139
Amul Cheese Slices	4.105
Amul Basundi	4.103
Amul Whey Protein	4.094
Amul Dark Chocolate	4.094
Amul Fresh Cream	4.093
Amul Taaza Milk	4.082
Amul Lassi	4.075
Amul Gulab Jamun Mix	4.063
Amul Kool Flavoured Milk	4.039
Amul Cheese Spread	4.031
Amul Gold Milk	4.019